A Counterexample to the High-Dimensional Slow-Epsilon Orthogonality Hope

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Source Question

The source paper is B. Burshteyn and A. Volberg, *Orthogonality in normed spaces*, arXiv:2107.02491. The paper defines X_1 to be ε-almost orthogonal to X_2 when every unit vector in X_1 has distance at least $1-\varepsilon$ to X_2.

At the end of Section 7, after constructing examples with no exactly orthogonal high-dimensional subspace, the authors ask whether one might still always find high-dimensional subspaces which are nearly orthogonal with a sequence of errors tending to zero:

$m=1$

In this space we can find the sequence F_m, E_m of finitely dimensional subspaces, such that $\dim F_m - \dim E_m = m$, and we can find a sequences of numbers $\{\varepsilon_m\}$, $\varepsilon_m \to 0$, such that for any subspace K_m of F_m, the orthogonality of K_m to E_m is *worse* than $1-\varepsilon_m$.

Open question. So, we cannot find *very orthogonal* K_m. But maybe this sequence $\{\varepsilon_m\}$ above goes to zero too fast? So, still maybe, given a space X and given F_m, E_m of finitely dimensional subspaces, such that $\dim F_m - \dim E_m = m$, we can always find *some* slowly going to zero sequence $\{\varepsilon_m\}$, such that K_m that is $1-\varepsilon_m$ orthogonal to E_m does exist. This we do not know.

8. Why that was counterintuitive, Grassmannians and their cohomology

The result of the previous Section seems counterintuitive. Let us explain why it seems counterintuitive.

There are elementary proofs of Borsuk–Ulam theorem, but most common proofs go through the following reasoning. A simple commutative diagram shows that if

The wording in that bookkeeping paragraph suppresses the dimension of K_m. Read literally without a dimension condition, the question is trivial in the positive direction: by the paper's Theorem 5.1, a one-dimensional exactly orthogonal subspace always exists whenever $\dim E < \dim F$. The surrounding discussion, however, is about high-dimensional subspaces, especially the case $\dim K_m = \dim F_m - \dim E_m = m$. This packet answers that natural high-dimensional version negatively.

Definitions

For subspaces $E \subset F$ of a normed space and a nonzero subspace $K \subset F$, set

$$\alpha(K, E; F) = \inf\{\operatorname{dist}_F(k, E) : k \in K,\ \|k\|_F = 1\}.$$

Thus K is ε-almost orthogonal to E exactly when $\alpha(K, E; F) \geq 1 - \varepsilon$.

We write $d(Y, Z)$ for Banach-Mazur distance between finite-dimensional normed spaces of the same dimension.

Proof Intuition

There are two competing facts. First, a quotient map detects orthogonality: if $q : F \to F/E$ and $\operatorname{dist}(k, E) \geq \alpha\|k\|$ on K, then $q|_K$ embeds K into the quotient with distortion at most $1/\alpha$.

Second, choose the quotient F_m/E_m to be uniformly Hilbertian while keeping F_m equal to $\ell_\infty^{O(m)}$. Then a very orthogonal high-dimensional K would make a high-dimensional subspace of $\ell_\infty^{O(m)}$ uniformly Hilbertian.

That is impossible. If an r-dimensional subspace of ℓ_∞^N were D-Euclidean, the N coordinate functionals would cover the Euclidean sphere by N pairs of caps of angular size about $1/D$. Spherical cap estimates force $N \geq \exp(cr/D^2)$, or $D \geq c\sqrt{r/\log N}$.

Lemma 1 (Euclidean subspaces of ℓ_∞^N are logarithmically small)**.** *There is a universal constant $c > 0$ such that, for every r-dimensional subspace $K \subset \ell_\infty^N$,*

$$d(K, \ell_2^r) \geq c\sqrt{\frac{r}{\log(2N)}}.$$

Proof. The assertion is trivial after decreasing c when the right hand side is less than 1, so suppose K is D-isomorphic to ℓ_2^r with $D \geq 1$. After rescaling the Euclidean norm $|\cdot|$ on K, we may assume

$$\|x\|_\infty \leq |x| \leq D\|x\|_\infty \qquad (x \in K).$$

Let $f_1, \ldots, f_N$ be the coordinate functionals restricted to K. The first inequality gives $\|f_i\|_{(K,|\cdot|)^*} \leq 1$. Hence $f_i(x) = \langle x, a_i\rangle$ for some a_i in the Euclidean unit ball. The second inequality implies that for every x on the Euclidean unit sphere S^{r-1},

$$\max_{1\leq i\leq N} |\langle x, a_i\rangle| = \|x\|_\infty \geq \frac{1}{D}.$$

Thus S^{r-1} is covered by the N sets

$$\{x \in S^{r-1} : |\langle x, a_i\rangle| \geq 1/D\}.$$

For each nonzero a_i, normalizing a_i only enlarges this set. The standard spherical cap estimate gives, for a universal $c_0 > 0$,

$$\sigma\{x \in S^{r-1} : |\langle x, u\rangle| \geq 1/D\} \leq 2\exp(-c_0 r/D^2)$$

for every unit vector u. Therefore $1 \leq 2N\exp(-c_0 r/D^2)$, whence $D \geq c\sqrt{r/\log(2N)}$ after changing constants. □

Theorem 1. *There are universal constants $A, C < \infty$, a Banach space X, and finite-dimensional subspaces $E_m \subset F_m \subset X$ such that*

$$\dim(F_m/E_m) = m$$

and every r-dimensional subspace $K \subset F_m$ satisfies

$$\alpha(K, E_m; X) \le C\sqrt{\frac{\log(Am)}{r}}.$$

In particular, for $r = m$,

$$\sup_{\dim K = m} \alpha(K, E_m; X) \le C\sqrt{\frac{\log(Am)}{m}} \longrightarrow 0.$$

Consequently there is no sequence $\varepsilon_m \to 0$ for which these pairs admit m-dimensional subspaces $K_m \subset F_m$ that are $1 - \varepsilon_m$ orthogonal to E_m.

Proof. We use Kashin's theorem in the following standard form: there are universal constants A and C_K such that for every m there is an integer $N_m \le Am$ and an m-dimensional subspace $G_m \subset \ell_1^{N_m}$ with

$$d(G_m, \ell_2^m) \le C_K.$$

Let $F_m = \ell_\infty^{N_m}$ and define

$$E_m = G_m^\perp = \{x \in \ell_\infty^{N_m} : g(x) = 0 \text{ for all } g \in G_m\},$$

where $\ell_1^{N_m}$ is identified with $(\ell_\infty^{N_m})^*$ in the usual coordinate duality. The quotient $Q_m = F_m/E_m$ is isometric to G_m^*. Since G_m is C_K-Hilbertian, so is G_m^*; hence

$$d(Q_m, \ell_2^m) \le C_K.$$

Embed the spaces F_m as disjoint coordinate blocks in $X = (\oplus_{m=1}^\infty F_m)_\infty$. For vectors in a single block, distances to E_m computed in X and in F_m agree.

Fix an r-dimensional subspace $K \subset F_m$ and put $\alpha = \alpha(K, E_m; F_m)$. If $\alpha = 0$, there is nothing to prove. Assume $\alpha > 0$, and let $q_m : F_m \to Q_m$ be the quotient map. For $k \in K$,

$$\alpha\|k\|_{F_m} \le \|q_m k\|_{Q_m} \le \|k\|_{F_m}.$$

Thus $q_m|_K$ is an isomorphism from K onto the r-dimensional subspace $q_m K \subset Q_m$ with distortion at most $1/\alpha$. Since every subspace of the C_K-Hilbertian space Q_m is again C_K-Hilbertian,

$$d(K, \ell_2^r) \le \frac{C_K}{\alpha}.$$

On the other hand, $K \subset F_m = \ell_\infty^{N_m}$, so Lemma 1 gives

$$d(K, \ell_2^r) \ge c\sqrt{\frac{r}{\log(2N_m)}}.$$

Combining these inequalities and using $N_m \le Am$ yields

$$\alpha \le \frac{C_K}{c}\sqrt{\frac{\log(2Am)}{r}},$$

which is the desired estimate after absorbing constants.

Taking $r = m$ makes the right hand side tend to zero. If $\varepsilon_m \to 0$, then $1 - \varepsilon_m \to 1$, so for all sufficiently large m no m-dimensional $K_m \subset F_m$ can satisfy $\alpha(K_m, E_m; X) \ge 1 - \varepsilon_m$. $\square$

Verification and Scope

The proof is analytic and uses no computation. The nontrivial external input is Kashin's theorem that $\ell_1^{O(m)}$ contains m-dimensional subspaces uniformly isomorphic to Hilbert space. The obstruction for subspaces of ℓ_∞^N is proved above from a spherical-cap estimate.

The result is stronger than just a negative answer for $\dim K_m = m$: it rules out any r_m-dimensional nearly orthogonal subspaces with $r_m/\log m \to \infty$. It does not contradict the one-dimensional Krein-Krasnoselski-Milman/Borsuk-Ulam theorem used in the source paper.

Novelty Check

The local run indexes were searched for arXiv:2107.02491, the title *Orthogonality in normed spaces*, and phrases including `slowly going to zero`, `K_m`, `Birkhoff-James`, and `finitely strictly singular lambda-adjusted`. No existing packet for this source question was found; only unrelated Birkhoff-James packets and targets appeared.

Web searches for exact phrases such as `"2107.02491" "slowly going to zero"`, `"Orthogonality in normed spaces" "K_m"`, and `"finitely strictly singular" "lambda-adjusted" "dim M_n"` found the source paper and related orthogonality papers, but no explicit later solution of the Section 7 high-dimensional slow-epsilon question. The proof should be reviewed for novelty against the local theory of Banach spaces: the ingredients are classical, but the assembly appears to directly answer the source's high-dimensional question.

References

[1] B. Burshteyn and A. Volberg, *Orthogonality in normed spaces*, arXiv:2107.02491, 2021.

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[3] T. Figiel, J. Lindenstrauss and V. D. Milman, *The dimension of almost spherical sections of convex bodies*, Acta Math. 139 (1977), 53–94.